# Syllabus Details

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|---|---|
| Syllabus ID: | 12035 |
| Syllabus Name: | **Machine Learning_Học máy** |
| Course Name English: | **Machine Learning** |
| Subject Code: | **AIL304m** |
| Learning-Teaching Method: | Blended,In-class lecture,Inquiry-based Teaching,Offline |
| NoCredit: | 3 |
| Degree Level: | Bachelor |
| Time Allocation: | 45h (60 sessions) contact hours + 1h final exam + 104h self-study |
| Pre-Requisite: | MAS291, MAE101 |
| Description: | In this course, student will build and train neural network, and you will learn the best practices to train and develop test sets and analyze bias/variance for building deep learning applications; be able to use standard neural network techniques such as initialization, L2 and dropout regularization, hyperparameter tuning, batch normalization, and gradient checking; implement and apply a variety of optimization algorithms, such as mini-batch gradient descent, Momentum, RMSprop and Adam, and check for their convergence; and implement a neural network in TensorFlow. Student make extensive use of tools from Linear Algebra, Calculus, Optimization. |
| StudentTasks: | - Students must attend at least 80% of contact slots in order to be accepted to the final examination.<br>- Student is responsible to do all exercises, assignments and labs given by instructor in class or at home and submit on time<br>- Use laptop in class only for learning purpose<br>- Promptly access to the https://flm.fpt.edu.vn/ for up-to-date course information |
| Tools: | - Jupyter notebook |
| Scoring Scale: | 10 |
| DecisionNo MM/dd/yyyy: | 1286/QĐ-ĐHFPT dated 11/22/2024 |
| IsApproved: | **True** |
| Note: | |
| Is Scored: | **True** |
| MinAvgMarkToPass: | 5 |
| IsActive: | True |
| ApprovedDate: | 11/22/2024 |

8 material(s)

| No. | Material Description | Author | Publisher | Published Date | Edition | ISBN | Is Main Material | Is Hard Copy | Is Online | Note |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | https://www.coursera.org/specializations/deep-learning | Coursera | Coursera | | | | True | False | True | |
| 2 | https://www.coursera.org/learn/neural-networks-deep-learning | Coursera | Coursera | | | | True | False | True | |
| 3 | https://www.coursera.org/learn/deep-neural-network?specialization=deep-learning | Coursera | Coursera | | | | True | False | True | |
| 4 | https://www.coursera.org/learn/machine-learning-projects?specialization=deep-learning | Coursera | Coursera | | | | True | False | True | |

| No. | Material Description | Author | Publisher | Published Date | Edition | ISBN | Is Main Material | Is Hard Copy | Is Online | Note |
|---|---|---|---|---|---|---|---|---|---|---|
| 5 | https://www.coursera.org/learn/machine-learning-linear-algebra?specialization=mathematics-for-machine-learning-and-data-science | Coursera | Coursera | | | | True | False | True | |
| 6 | https://www.coursera.org/learn/tensorflow-for-ai-get-to-know-tensorflow/ungradedLti/fsCCK/tensorflow-for-ai-get-to-know-tensorflow | Coursera | Coursera | | | | True | False | True | Project 1 |
| 7 | https://www.coursera.org/projects/googlecloud-classify-images-of-cats-and-dogs-using-transfer-learning-6qgew | Coursera | Coursera | | | | True | False | True | Project 2 |
| 8 | https://www.coursera.org/projects/introduction-to-fastapi-framework | Coursera | Coursera | | | | True | False | True | Project 3 |

5 LO(s)

| No. | CLO Name | CLO Details |
|---|---|---|
| 1 | CLO1 | Deeply understand the math that makes machine learning algorithms work. |
| 2 | CLO2 | Able to build, train and apply fully connected deep neural networks, implement efficient neural networks, identify key parameters in a neural network's architecture, and apply deep learning to students' own applications. |
| 3 | CLO3 | Able to train and develop test sets, analyze bias/variance for building deep learning applications, apply optimization algorithms, and implement a neural network using TensorFlow. |
| 4 | CLO4 | Understand complex ML settings, diagnose errors in a machine learning system, prioritize strategies for reducing errors, and apply end-to-end learning, transfer learning, and multi-task learning. |
| 5 | CLO5 | Practical project with Tensorflow |

View mapping of CLOs to PLOs

Download All Student Material

| Session | Topic | Learning-Teaching Type | LO | ITU | Student Materials | S-Download | Student's Tasks | URLs |
|---|---|---|---|---|---|---|---|---|
| 1 | Math Introduction<br>1.1: Systems of Linear equations: two variables. | Offline | CLO1 | IT | Slide, Videos of Systems of Linear equations | AIL304m | Read Slide and Watch Video | |
| 2 | Math Introduction<br>1.2: Systems of Linear Equations: three variables | Offline | CLO1 | IT | Slide, Videos of Systems of Linear equations | | Read Slide and Watch Video | |
| 3 | Solving systems of Linear Equations<br>1.3: Solving systems of Linear Equations: Elimination | Offline | CLO1 | IT | Slide, Videos of Solving systems of Linear Equations | | Read Slide and Watch Video | |
| 4 | 1.4: Solving systems of Linear Equations: Row Echelon Form and Rank | Offline | CLO1 | IT | Slide, Videos of Solving systems of Linear Equations | | Read Slide and Watch Video | |
| 5 | Vectors and Linear Transformations<br>1.5 Vectors | Offline | CLO1 | IT | Slide, Videos of Vectors and Linear Transformations | | Read Slide and Watch Video | |

| Session | Topic | Learning-Teaching Type | LO | ITU | Student Materials | S-Download | Student's Tasks | URLs |
|---|---|---|---|---|---|---|---|---|
| 6 | 1.6: Linear transformations | Offline | CLO1 | IT | Slide, Videos of Vectors and Linear Transformations | | Read Slide and Watch Video | |
| 7 | Programming Assignment: Single Perceptron Neural Networks for Linear Regression<br>Reset your deadline to | Offline | CLO1 | U | Exercise 1 file | | Do Exercise 1 | |
| 8 | Determinants and Eigenvectors<br>1.7: Determinants In-depth<br>1.8: Eigenvalues and Eigenvectors | Offline | CLO1 | IT | Slide, Videos of Determinants and Eigenvectors | | Read Slide and Watch Video | |
| 9 | Programming Assignment: Eigenvalues and Eigenvectors | Offline | CLO1 | U | Exercise 2 file | | Do Exercise 2 | |
| 10 | Functions of one variable: Derivative and optimization<br>1.9: Derivatives | Offline | CLO1 | IT | Slide, Videos of Functions of one variable: Derivative and optimization | | Read Slide and Watch Video | |
| 11 | Functions of one variable: Derivative and optimization<br>1.10: Optimization with derivatives | Offline | CLO1 | IT | Slide, Videos of Functions of one variable: Derivative and optimization | | Read Slide and Watch Video | |
| 12 | Programming Assignment: Optimizing Functions of One Variable: Cost Minimization | Offline | CLO1 | U | Exercise 3 file | | Do Exercise 3 | |
| 13 | Functions of two or more variables: Gradients and gradient descent<br>1.11: Gradients and optimization<br>1.12: Gradient Descent | Offline | CLO1 | IT | Slide, Videos of Functions of two or more variables: Gradients and gradient descent | | Read Slide and Watch Video | |
| 14 | Functions of two or more variables: Gradients and gradient descent<br>1.13: Optimization in Neural Networks | Offline | CLO1 | IT | Slide, Videos of Functions of two or more variables: Gradients and gradient descent | | Read Slide and Watch Video | |
| 15 | Functions of two or more variables: Gradients and gradient descent<br>1.14:Beyond Gradient Descent: Newton's Method | Offline | CLO1 | IT | Slide, Videos of Functions of two or more variables: Gradients and gradient descent | | Read Slide and Watch Video | |
| 16 | Lab01: Regression with Perceptron | Offline | CLO1 | U | Lab 1 file | | Do Lab 1 | |
| 17 | Review Lab01 | Offline | CLO1 | U | Lab 1 file | | | |
| 18 | Progess test 1 | Offline | CLO1 | U | - Content Module 1. | | Do progress test 1 | |
| 19 | 2. Introduction to Artificial Intelligence and Machine Learning | Offline | CLO2 | IT | Slide , Videos of Introduction to Artificial Intelligence and Machine Learning | | Read Slide and Watch Video | |

| Session | Topic | Learning-Teaching Type | LO | ITU | Student Materials | S-Download | Student's Tasks | URLs |
|---|---|---|---|---|---|---|---|---|
| 20 | 2.1 Neural Networks Basics | Offline | CLO2 | T | Slide, Videos of Neural Networks Basics | | Read Slide and Watch Video | |
| 21 | 2.2 Logistic Regression as a Neural Network | Offline | CLO2 | T | Slide, Videos of Neural Networks Basics | | Read Slide and Watch Video | |
| 22 | 2.3 Python and Vectorization | Offline | CLO2 | T | Slide, Videos of Neural Networks Basics | | Read Slide and Watch Video | |
| 23 | Programming Assignments: Logistic Regression with a Neural network mindset | Offline | CLO2 | U | Exercise 4 file | | Do Exercise 4 file | |
| 24 | 2.4 Shallow neural networks | Offline | CLO1 | T | Slide, Videos of Shallow neural networks | | Read Slide and Watch Video | |
| 25 | 2.4 Shallow neural networks | Offline | CLO1 | T | 0 | | Read Slide and Watch Video | |
| 26 | Programming Assignments:Planar Data Classification with One Hidden Layer | Offline | CLO2 | U | Exercise 5 file | | Do Exercise 5 | |
| 27 | Guidedline project 01: TensorFlow for AI: Get to Know Tensorflow | Offline | CLO4,CLO5 | U | Link in the Materials | | View in the link | |
| 28 | Guidedline project 01: TensorFlow for AI: Get to Know Tensorflow | Offline | CLO4,CLO5 | U | Link in the Materials | | View in the link | |
| 29 | Guidedline project 01: TensorFlow for AI: Get to Know Tensorflow | Offline | CLO4,CLO5 | U | Link in the Materials | | View in the link | |
| 30 | 2.5 Deep Neural Networks | Offline | CLO2 | T | Slide, Videos of Deep Neural Networks | | Read Slide and Watch Video | |
| 31 | 2.5 Deep Neural Networks | Offline | CLO2 | T | Slide, Videos of Deep Neural Networks | | Read Slide and Watch Video | |
| 32 | 2.5 Deep Neural Networks | Offline | CLO2 | T | Slide, Videos of Deep Neural Networks | | Read Slide and Watch Video | |
| 33 | Review Project 1 | Offline | CLO2 | U | Link in the Materials | | View in the link | |
| 34 | 2.5 Deep Neural Networks | Offline | CLO2 | T | Slide, Videos of Deep Neural Networks | | Read Slide and Watch Video | |
| 35 | 2.5 Deep Neural Networks | Offline | CLO2 | T | Slide, Videos of Deep Neural Networks | | Read Slide and Watch Video | |

| Session | Topic | Learning-Teaching Type | LO | ITU | Student Materials | S-Download | Student's Tasks | URLs |
|---|---|---|---|---|---|---|---|---|
| 36 | 2.5 Deep Neural Networks | Offline | CLO2 | T | Slide, Videos of Deep Neural Networks | | Read Slide and Watch Video | |
| 37 | Guidedline project 2: Neural Network from Scratch in TensorFlow | Offline | CLO5 | U | Link in the Materials | | View in the link | |
| 38 | Guidedline project 2: Neural Network from Scratch in TensorFlow | Offline | CLO5 | U | Link in the Materials | | View in the link | |
| 39 | Guidedline project 2: Neural Network from Scratch in TensorFlow | Offline | CLO5 | U | Link in the Materials | | View in the link | |
| 40 | Lab 2 (Neural Networks) | Offline | CLO2 | U | Lab 2 file | | Do Lab 2 | |
| 41 | Lab 2 (Neural Networks) | Offline | CLO2 | U | Lab 2 file | | Do Lab 2 | |
| 42 | Lab 2 (Neural Networks) | Offline | CLO2 | U | Lab 2 file | | Do Lab 2 | |
| 43 | 3.1 Practical aspects of Deep Learning | Offline | CLO,3 | T | Slide, Videos of Practical aspects of Deep Learning | | Read Slide and Watch Video | |
| 44 | 3.1 Practical aspects of Deep Learning | Offline | CLO3 | T | Slide, Videos of Practical aspects of Deep Learning | | Read Slide and Watch Video | |
| 45 | 3.2 Optimization algorithms | Offline | CLO3 | T | Slide, Videos of Optimization algorithms | | Read Slide and Watch Video | |
| 46 | 3.2 Optimization algorithms | Offline | CLO3 | T | Slide, Videos of Optimization algorithms | | Read Slide and Watch Video | |
| 47 | 3.2 Optimization algorithms | Offline | CLO3 | T | Slide, Videos of Optimization algorithms | | Read Slide and Watch Video | |
| 48 | Review Lab 2 | Offline | CLO2 | U | Lab 2 file | | | |
| 49 | Review project 2 | Offline | CLO5 | U | Link in the Materials | | View in the link | |
| 50 | 3.3 Hyperparameter tuning, Batch Normalization and Programming Frameworks | Offline | CLO3 | T | Slide, Videos of Hyperparameter tuning, Batch Normalization and Programming Frameworks | | Read Slide and Watch Video | |
| 51 | 3.3 Hyperparameter tuning, Batch Normalization and Programming Frameworks | Offline | CLO3 | T | Slide, Videos of Hyperparameter tuning, Batch Normalization and Programming Frameworks | | Read Slide and Watch Video | |
| 52 | 3.3 Hyperparameter tuning, Batch Normalization and Programming Frameworks | Offline | CLO3 | T | Slide, Videos of Hyperparameter tuning, Batch Normalization and Programming Frameworks | | Read Slide and Watch Video | |
| 53 | 3.3 Hyperparameter tuning, Batch Normalization and Programming Frameworks | Offline | CLO3 | T | Slide, Videos of Hyperparameter tuning, Batch Normalization and Programming Frameworks | | Read Slide and Watch Video | |

| Session | Topic | Learning-Teaching Type | LO | ITU | Student Materials | S-Download | Student's Tasks | URLs |
|---|---|---|---|---|---|---|---|---|
| 54 | Progess test 2 | Offline | CLO2, CLO3 | U | - Content Module 2 and 3. | | Do Progress test 2 | |
| 55 | 4.1 ML Strategy 1 | Offline | CLO4 | T | Slide, Videos of ML Strategy 1 | | Read Slide and Watch Video | |
| 56 | 4.1 ML Strategy 1 | Offline | CLO4 | T | Slide, Videos of ML Strategy 1 | | Read Slide and Watch Video | |
| 57 | 4.1 ML Strategy 1<br>4.2 ML Strategy 2 | Offline | CLO4 | T | Slide, Videos of ML Strategy 1 and ML Strategy 2 | | Read Slide and Watch Video | |
| 58 | Guidedline project 3: Deploy Models with TensorFlow Serving and Flask | Offline | CLO5 | U | Link in the Materials | | View in the link | |
| 59 | Guidedline project 3: Deploy Models with TensorFlow Serving and Flask | Offline | CLO5 | U | Link in the Materials | | View in the link | |
| 60 | Review project 3 | Offline | CLO5 | U | Link in the Materials | | View in the link | |

0 Constructive question(s)

4 assessment(s)

| No. | Category | Type | Part | Weight | Completion Criteria | Duration | CLO | Question Type | No Question | Knowledge and Skill | Grading Guide | Note |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | Lab | on-going | 2 | 20.0% | >0 | 1 day | Lab1: CLO1<br>Lab 2: CLO2 | N/A | 2 | Lab1: cover content of Module 1<br>Lab 2: cover content of Module 2 | at home, by instructor | Instructor has resposibility to review the test for students after graded. |

| No. | Category | Type | Part | Weight | Completion Criteria | Duration | CLO | Question Type | No Question | Knowledge and Skill | Grading Guide | Note |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | Progress test | on-going | 2 | 20.0% | >0 | 20'-30' | PT1: CLO1; PT2: CLO2, CLO3, CLO4 | Multiple choices Marked by Computer or a suitable format | 20 | PT1: cover content of Module 1; PT2: cover content of Module 2,3,4 | in class, by instructor | Instruction and shedules for Progress tests must be presented in the Course Implementation Plan approved by director of the campus.<br><br>Progress test must be taken right after the last lectures of required material.<br><br>Instructor has resposibility to review the test for students after graded. |
| 3 | Project | on-going | 3 | 30.0% | >0 | 2 weeks | Project 1, 2: CLO2 Project 3: CLO4 | N/A | 2 | Project 1, 2: cover content of Module 2 Project 3: cover content of Module 4 | at home, by instructor | Instructor has resposibility to review the test for students after graded. |
| 4 | Final exam | Final exam | 1 | 30.0% | 4 | 60' | CLO1, CLO2, CLO3, CLO4, | Multiple choices Marked by Computer | 50 | all studied modules | by exam board, using computer | The exam questions must be updated or different at least 70% to the previous ones. |